

Current Position

Postdoctoral Research Fellow Carleton University 2025–

Education

Ph.D University of Toronto 2020–2026
M.Sc. Concordia University 2017–2019
B.Sc (Hons) University of Toronto 2013–2017

Publications

In Preparation

****Goldman JA****, Fortin MJ, James PMA. Integrating data from multiple sources to better understand the interaction between the spatial legacies of spruce budworm outbreak and wildfire burn severity in the boreal forest.

Millard K, Pelletier N, Thompson DK, ****Goldman JA****, Pontone N, Guindon L. Severe droughts are putting Canadian peatland carbon stocks at risk: an analysis of fire selectivity and behavior in peatlands over the extreme 2023 fire season.

Wilkinson SL, ****Goldman JA****, Krieger-Pottruff E, James PMA.

****Goldman JA****. Combining propensity score matching and causal forests to evaluate the spatially variable effects of spruce budworm defoliation on wildfire burn severity.

In Review

Goldman JA, Fortin MJ, James PMA. Within and among year moisture availability influences wildfire burn severity in the boreal forest. *Fire Ecology*.

Published

7. Goldman JA, Fortin MJ, James PMA. 2025. Snow dynamics and forest structure interact to increase wildfire burn severity in the boreal forest. *Forest Ecology and Management*.
6. Messaoud Y, Reid A, Tchebakova NM, Goldman JA, & Hofgaard A. 2022. The historical complexity of tree height growth dynamic associated with climate change in western north america. *Forests*, 13(5): 738.
5. Crane AL, Bairos-Novak KR, Goldman JA, Brown GE. 2022. Chemical disturbance cues in aquatic systems: a review and prospectus. *Ecological Monographs*, 92(1): e01487.
4. Goldman JA, Crane AL, Feyten LEA, Collins E, Brown GE. 2022. Disturbance cue communication is shaped by emitter diet and receiver background risk in Trinidadian guppies. *Current Zoology*, 68(4): 433440.
3. Brown GE, Demers EM, Goldman JA, Singh A, Chivers DP, Ferrari MCO. 2020. Unpredictable risk enhances induced neophobia in northern red-bellied dace. *Animal Behaviour*, 168: 121127.
2. Goldman JA, Desormeaux IS, Brown GE. 2020. Disturbance cues function as sources of risk assessment information under natural conditions. *Freshwater Biology*, 65(5): 981986.
1. Goldman JA, Feyten LEA, Ramnarine IW, Brown GE. 2020. Sender and receiver experience with risk alters the response to disturbance cues by Trinidadian guppies. *Current Zoology*, 66(3): 255261.

Presentations

- 2024 Comparing forest recovery trajectories following wildfire in the presence and absence of spruce budworm defoliation using remotely sensed data. IALE-North America Annual Meeting, Oklahoma City, Oklahoma.
- 2023 The effects of snowmelt timing on wildfire burn severity in the boreal forest. IALE-North America Annual Meeting, Riverside, California.

- 2022 Time-lagged and cumulative effects of moisture availability on burn severity in the boreal forest. Wildland Fire Canada Conference, Edmonton, AB.
- 2022 Trends in burn severity across an east-west gradient. Ecological Society of America Annual Meeting, Montreal, QC.
- 2022 Spatial and temporal trends in landscape patterns of burn severity in the boreal shield ecozone of Ontario. North American Forest Ecology Workshop (virtual).

Scholarships & Awards

- 20242025 School of Graduate Studies Doctoral Completion Award
- 20232024 Queen Elizabeth II Graduate Scholarship in Science and Technology
- 2022 North American Forest Ecology Workshop Student Presentation Award 3rd Place
- 20202021 Edward H. Buckley and Grace B. Buckley Graduate Scholarship in Forestry
- 20202021 GSEF Faculty of Forestry Award
- 20202021 Louis J. Nozzollilo Fellowship in Forestry Research
- 20202023 NSERC Postgraduate Scholarship Doctoral

Teaching

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| 2023 | FOR3012: Analytical Methods in Forestry Statistics (graduate)
Teaching Assistant | University of Toronto |
| 2020 | BIO120: Adaptation and Biodiversity
Teaching Assistant | University of Toronto |
| 2019 | BIOL 4259/9915: Research Hypothesis Testing (undergrad/grad)
Teaching Assistant | Concordia University |
| 2018 | BIOL 227: Laboratory Studies of Organisms
Teaching Assistant | Concordia University |
| 2017 | BIOL 201: Introductory Biology
Teaching Assistant | Concordia University |

Lectures & Workshops

- 2025 Harnessing The Power of Google Earth Engine with Python
- 2023 Multivariate Analysis and Ordination FOR3012

Professional Development

- 2023 Short Course on Forecasting for Decision-Making: An Epidemiological & Ecological Perspective (Invited)
- 2023 Structural Equation Modelling Summer School

Skills & Tools

- Languages & Environments R + RStudio, JavaScript + Google Earth Engine, Python + JupyterLab
- Workflows & Collaboration git + GitHub, project-oriented workflows (RProjects), RMarkdown, Quarto, Targets, Slack, PostgreSQL

Statistical Methods	Structural equation modelling, machine learning (XGBoost), spatial modelling, mixed effects modelling, multivariate analysis (ordination)
Remote Sensing Methods	Image acquisition, advanced image processing, supervised classification, spectral unmixing (Google Earth Engine), spectral indices